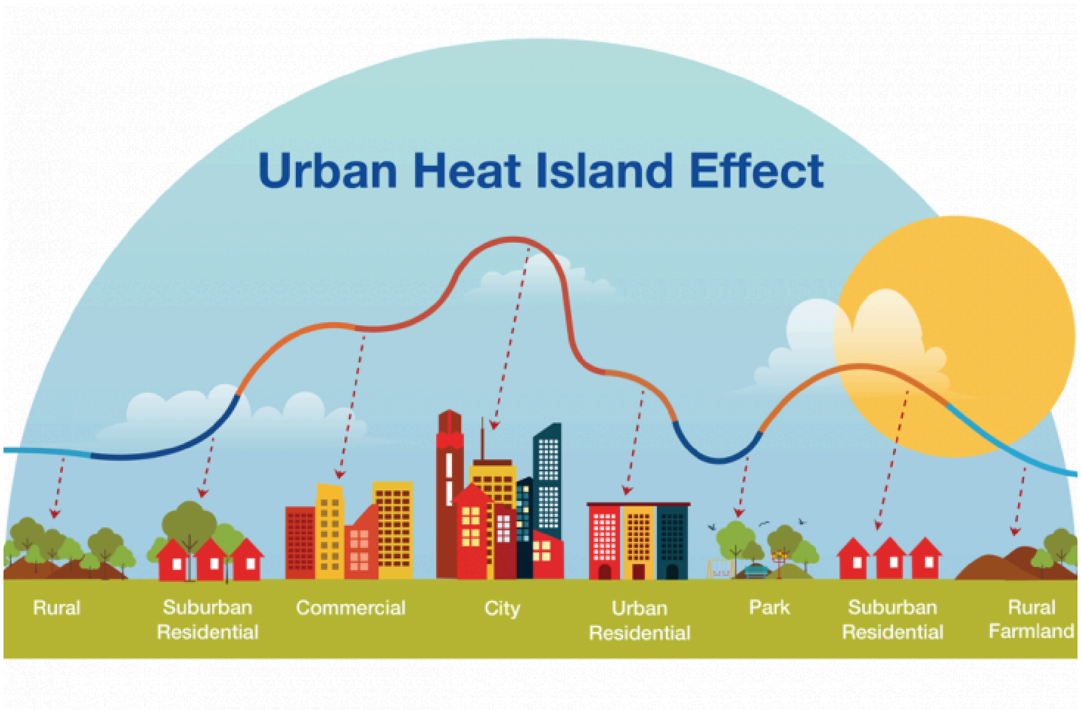

Understanding Urban Heat Through the Modeling of Temperature and Population

04.27.2026

— Data Science 2 Final Project Report

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Problem Statement

The central phenomenon of this study is the urban heat island effect, in which industrialized and densely developed cities experience consistently higher temperatures than their surrounding rural areas due to factors such as reduced vegetation, increased impervious surfaces, and heat generated from human activity (see Figure 1). The primary objective is to investigate how population size influences both the magnitude and variability of temperature fluctuations over time, with particular attention to how rapid urbanization, population density, and land-use changes amplify or dampen thermal behavior within cities.

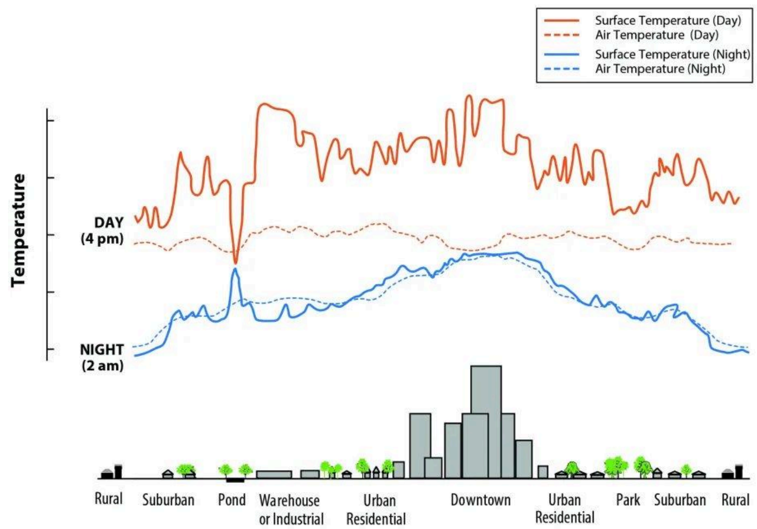

Figure 1: Variation of Temperature with Population Density

To address this, predictive models will be developed to characterize and forecast temperature patterns by integrating demographic data (e.g., population size and density) with spatial and environmental variables such as land cover, infrastructure distribution, and geographic location. These models may incorporate statistical and machine learning approaches to capture both temporal trends and spatial heterogeneity. The expected impact of this work is to provide data-driven insights for urban planners and policymakers, enabling more informed decisions on city design, green infrastructure implementation, and environmental regulations to mitigate heat accumulation and improve long-term urban sustainability.

Collection and Description of Datasets

Dataset 1: Global Land Temperatures by Major Cities

This dataset contains monthly average temperature records for major cities worldwide. The primary variable of interest is average temperature (dependent variable), along with supporting variables including date, city, latitude, and longitude. This dataset provides the temporal and spatial foundation for analyzing long-term climate trends (see Figure 2).

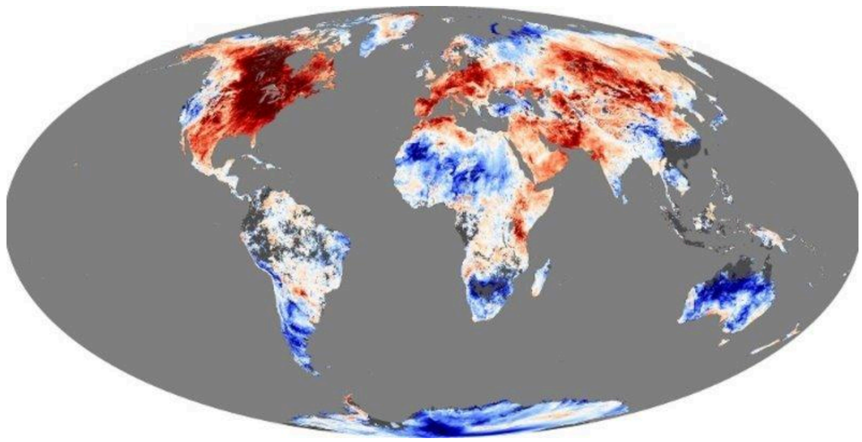

Figure 2: Visualization of Global Land Temperatures by Major Cities

Dataset 2: World Cities Population Database

This dataset includes city population data and country identifiers, which are used to quantify the level of urbanization. It enables the analysis of how population size and urban growth may relate to temperature patterns (see Figure 3).

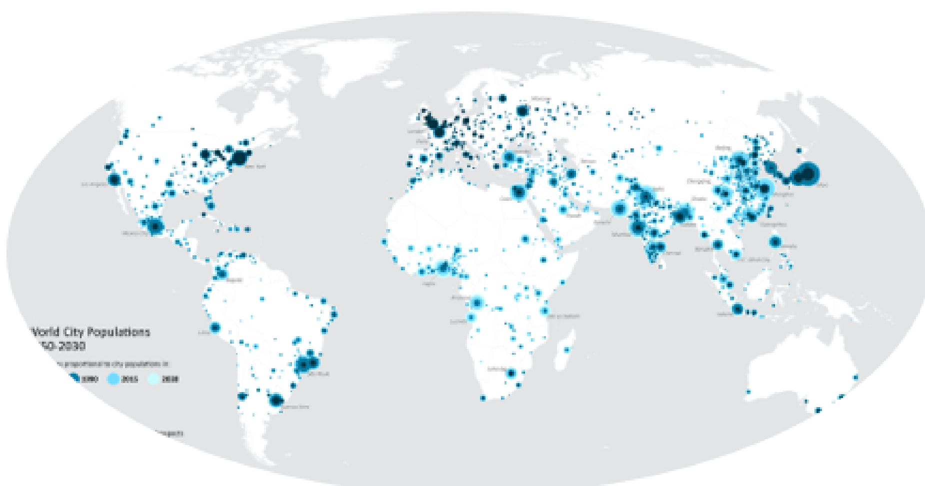

Figure 3: Visualization of Global Population Density

Data Combination

The two datasets were merged using common city and country identifiers to create a unified dataset linking temperature and urbanization metrics.

- Merged dataset size: 228,175 monthly records
- Number of cities: 100
- Countries: 49
- Time span: 1743 to 2013
- Population Range Used For Matching: 1,200,000 to 22,315,474
- Mean average temperature: 18.13

This combined dataset allows for integrated analysis of climate trends and urbanization effects over time. This effect is best summarized by analyzing the hottest and coldest cities.

Table 1: Five Warmest Cities by Mean Temperature

City	Country	Average Temperature (℃)	Population
Umm Durman	Sudan	29.08	1,200,000
Madras	India	28.42	4,681,087
Jiddah	Saudi Arabia	27.69	2,867,446
Ho Chi Minh	Vietnam	27.19	8,993,082
Bangkok	Thailand	27.16	5,104,476

Table 2: Five Coolest Cities by Mean Temperature

City	Country	Average Temperature (℃)	Population
Harbin	China	3.63	5,242,897
Saint Petersburg	Russia	3.92	5,351,935
Moscow	Russia	4.00	10,381,222
Montreal	Canada	4.45	1,762,949
Changchun	China	4.92	4,714,996

Tables 1 and 2 highlight a strong geographic clustering effect in global temperature distributions. The warmest cities are concentrated in regions close to the equator or within tropical/desert climates, such as Sudan, India, Saudi Arabia, Vietnam, and Thailand. These locations experience consistently high solar exposure throughout the year, leading to elevated mean temperatures with relatively low seasonal variation. In contrast, the coolest cities are primarily located in higher latitudes within the Northern Hemisphere, including China, Russia, and Canada. These regions experience harsher winters and lower annual solar input, resulting in significantly lower average temperatures. This reinforces earlier findings that latitude is a dominant driver of temperature, far more influential than other factors like population.

Additionally, these tables further support the observation that population size is not a strong determinant of average temperature. Both warm and cool city lists include highly populated urban centers (e.g., Bangkok and Moscow) as well as moderately sized cities, indicating no consistent relationship between city size and temperature extremes. Instead, temperature differences are largely governed by climate zone and geographic positioning, rather than urbanization level alone.

Data Preprocessing Techniques Applied

Before building any models, a critical preprocessing stage was performed to ensure the datasets were consistent, accurate, and suitable for analysis. One of the primary concerns was the presence of missing values within the temperature records. These gaps were handled by removing incomplete observations (rows) from the dataset. While this approach reduces the total number of data points, it preserves the integrity of the time series and avoids introducing bias or uncertainty through imputation. After this cleaning step, the final dataset consisted of 228,175 observations, which were subsequently split into 163,768 training samples and 64,407 testing samples for model development and evaluation.

Next, attention was given to temporal variables to better support time-based analysis. The original date column was transformed into a more usable numeric format by extracting year and month components. This allowed for clearer identification of seasonal patterns, cyclical behavior, and long-term temperature trends across different cities. By structuring time in this way, the dataset became more compatible with both statistical analysis and machine learning models that rely on explicit temporal features.

Several transformations were also required to properly handle categorical and numeric variables. The population field, originally stored as a string with comma separators (e.g., “1,000,000”), was reformatted by removing commas so it could be interpreted as a numeric variable. Additionally, geographic coordinates, initially stored as strings such as “40.7N” or “74.0W”, were converted into standard numerical values. This included assigning the correct sign based on cardinal direction (positive for north/east, negative for south/west), enabling accurate spatial analysis and compatibility with modeling tools.

Finally, in preparation for merging the temperature and population datasets, key identifiers such as city and country names were standardized. This involved applying uniform capitalization and removing any special or inconsistent characters that could interfere with matching records across datasets. This step was essential to ensure a reliable and accurate data fusion process, ultimately allowing both datasets to be combined seamlessly for integrated analysis of urbanization and temperature trends.

Visual Examination

The time-series visualization (Figure 4) reveals a clear long-term warming trend in mean temperatures across major cities. From the mid-18th century to the early 19th century, temperatures appear relatively volatile, with noticeable fluctuations and occasional sharp drops, likely reflecting limited data coverage or measurement inconsistencies in earlier records. However, beginning in the late 1800s and continuing through the 20th and early 21st centuries, the trend becomes more stable and steadily increasing. This gradual rise suggests a persistent upward shift in global urban temperatures over time, aligning with broader observations of climate warming. The reduced variability in later years also indicates improved data reliability and more consistent measurement practices.

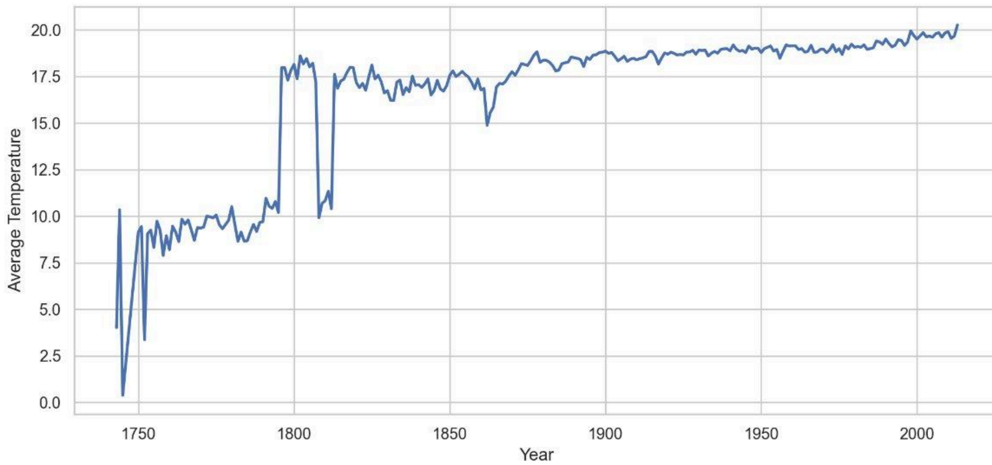

Figure 4: Mean Major-City Temperature vs Time

The monthly distribution plot (Figure 5) highlights strong seasonal patterns in temperature across cities. Warmer months (June through August) show higher median temperatures with relatively tighter interquartile ranges, indicating more consistent warm conditions globally. In contrast, winter months (December through February) exhibit lower medians and significantly wider spreads, reflecting greater variability in colder climates. The presence of outliers, particularly in summer and transitional months, suggests occasional extreme temperature events. Overall, this distribution confirms the expected cyclical nature of temperature while also emphasizing that variability is not uniform throughout the year, with colder months demonstrating greater dispersion across different geographic regions.

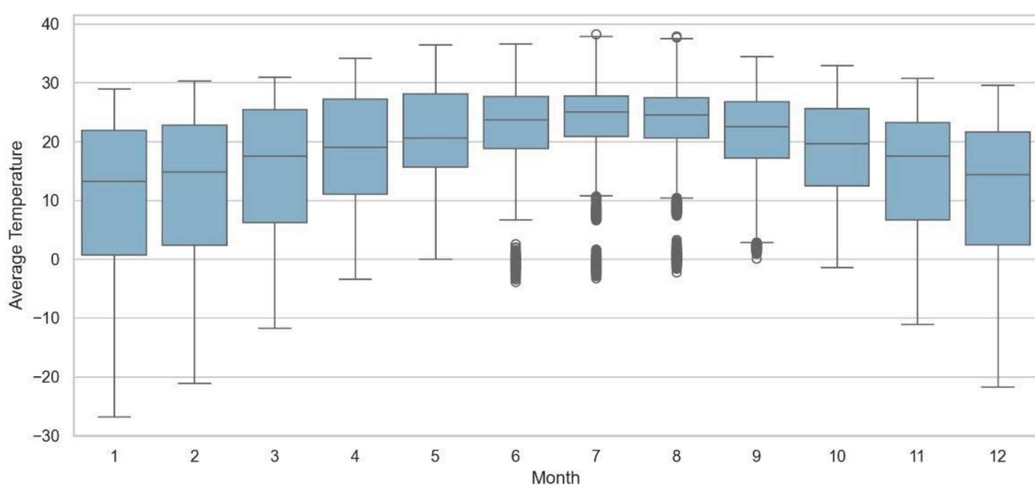

Figure 5: Monthly Temperature Distribution

The correlation heatmap (Figure 6) provides several important insights into the relationships between variables. Most notably, average temperature shows a moderate negative correlation with latitude (-0.39), confirming that temperatures generally decrease as one moves away from the equator. There is also a slight positive correlation between temperature and year (0.15), reinforcing the gradual warming trend observed earlier in the time-series analysis. Population and log population are, as expected, highly correlated (0.95), indicating that the logarithmic transformation preserves the underlying scale while improving numerical stability for modeling. Interestingly, temperature has almost no correlation with population, suggesting that urban size alone is not a strong direct predictor of average temperature at this scale. Another key observation is the strong negative correlation between temperature uncertainty and year (-0.75), indicating that measurement accuracy has improved significantly over time

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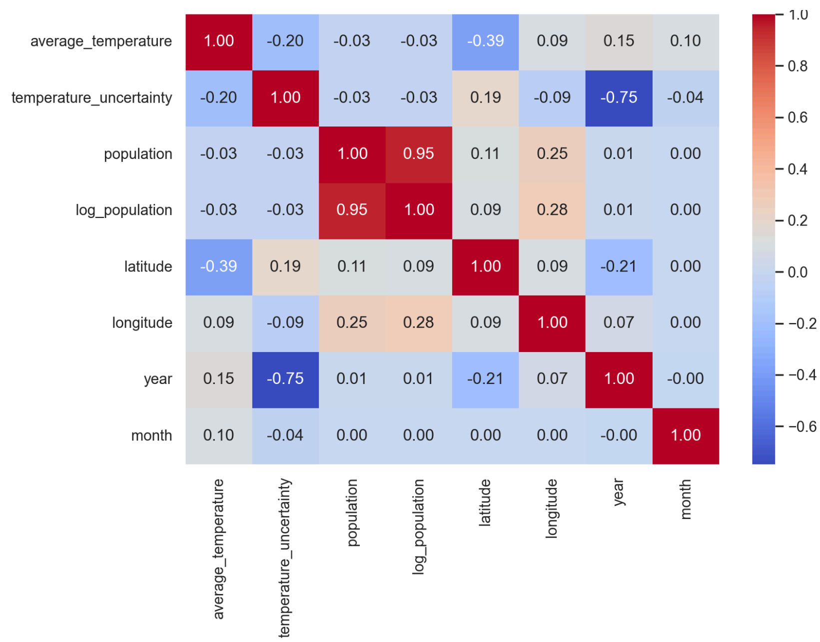

Figure 6: Correlation Heatmap of Population Dataset

The scatter plot of average temperature versus latitude (Figure 7) visually reinforces the relationship identified in the heatmap. There is a clear downward trend in temperature as latitude increases, with warmer temperatures concentrated near lower latitudes and colder, more variable temperatures observed at higher latitudes. The spread of data also increases at higher latitudes, reflecting greater seasonal variability and more extreme temperature ranges in those regions. Additionally, the color gradient (representing time) suggests a subtle upward shift in temperatures across many latitudes in more recent years, further supporting the presence of global warming trends. Overall, this plot highlights latitude as one of the most influential geographic factors affecting temperature distribution.

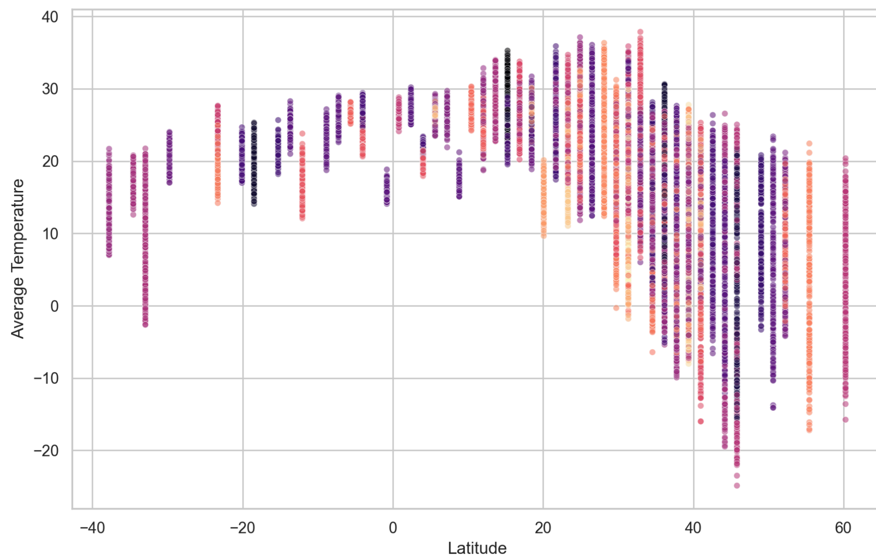

Figure 7: Scatter plot of Average Temperature vs Latitude

The scatter plot of average temperature versus log population (Figure 8) shows little to no clear linear relationship between urban population size and temperature. Data points are widely dispersed across all population levels, indicating that cities with both small and large populations experience a broad range of temperatures. This aligns with the earlier correlation analysis, where the population feature exhibited near-zero correlation with temperature. While there are clusters at certain population ranges (due to repeated city observations over time), there is no evident trend suggesting that increased urban population directly corresponds to higher average temperatures at a global scale.

However, the color gradient (representing time) suggests a subtle temporal effect embedded within the data. More recent observations tend to appear slightly higher in temperature across nearly all population levels, reinforcing the presence of a global warming trend rather than a population-driven one. Additionally, the large vertical spread at most population values indicates that geographic factors (such as latitude) play a significantly stronger role in determining temperature than urban size. Overall, this figure supports the conclusion that while urbanization may have localized effects, it is not a dominant predictor of average temperature when viewed across diverse global cities.

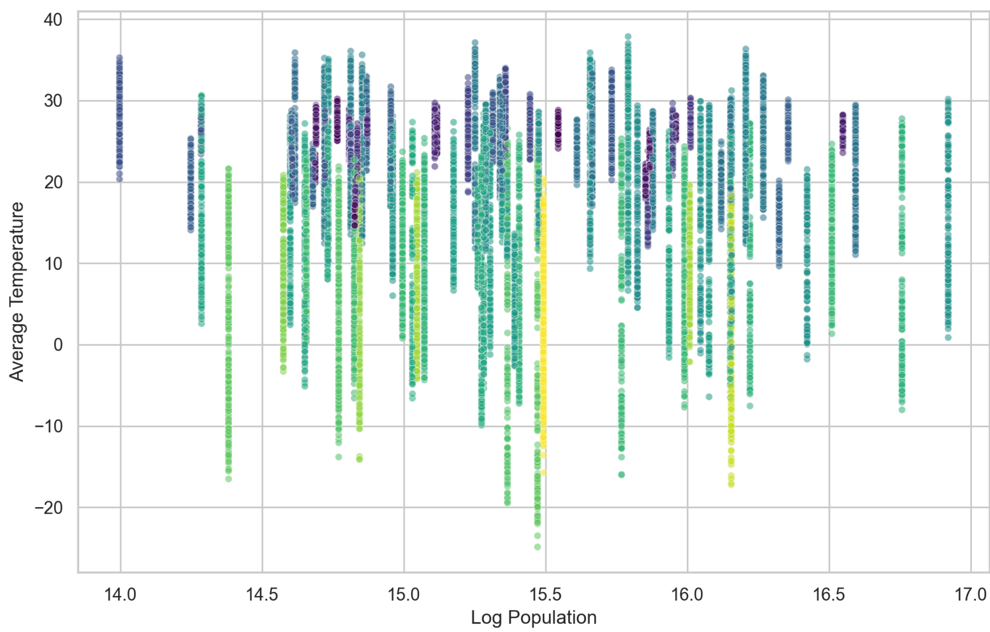

Figure 8: Average Temperature vs Log Transformed Temperature

Data Preprocessing Techniques Applied - Post EDA

Several preprocessing and feature engineering steps were applied to improve model performance and ensure meaningful interpretation of the data. The population variable was log-transformed due to its heavy right skew, as the dataset is dominated by a few extremely large cities. Applying the logarithmic transformation reduced this skewness and stabilized variance, allowing the model to better capture relationships without being disproportionately influenced by outliers. Additionally, month was transformed using sine and cosine functions to encode its cyclical nature. Since temperature follows a seasonal pattern, this transformation preserves continuity between December and January and enables the model to better learn periodic behavior.

To maintain the integrity of temporal analysis, the dataset was split into training and testing sets chronologically rather than randomly. This approach prevents data leakage, where information from future time periods could influence the model during training, and more accurately mimics a real-world forecasting scenario. In terms of data cleaning, latitude and longitude values were parsed from directional strings into signed numerical values, and the date field was decomposed into numeric components such as year and month. Furthermore, city and country names were standardized through consistent capitalization and removal of irregular characters to ensure proper merging of the temperature and population datasets.

Missing values were handled by removing incomplete observations instead of applying imputation. This decision was made because imputing temperature values without sufficient geographic or temporal context could introduce artificial patterns and noise into the dataset. By retaining only complete records, the analysis prioritizes data reliability over dataset size. Feature selection was then performed using a time series cross-validation approach, where subsets of features were evaluated based on their impact on model performance using RMSE as the ranking metric. Interestingly, the best-performing subset retained all candidate features, suggesting that each contributed some predictive value.

The exploratory analysis further reinforced several important insights. For example, average temperature showed a stronger relationship with latitude (-0.39) than with population (-0.03), indicating that geographic location plays a significantly larger role in determining temperature than urban size. The year variable exhibited a mild positive correlation with temperature, supporting the presence of a long-term warming trend consistent with global climate research. Additionally, temperature uncertainty displayed a strong negative correlation with year (-0.75), reflecting improvements in measurement accuracy over time as observational tools and methodologies have advanced. These findings not only validated expected physical relationships but also highlighted patterns beyond the original scope of the study.

Modeling Techniques Chosen

A range of models with increasing complexity was implemented to systematically evaluate predictive performance and understand the underlying relationships in the data. Reference (null) models were first constructed to predict average temperature using simple statistics such as the global mean or time-based averages. These models serve as a baseline, allowing us to quantify whether more sophisticated approaches meaningfully improve predictive accuracy.

Next, simple models such as multiple linear regression were applied to capture direct relationships between key features, including latitude, population (log-transformed), and temporal variables. These models assume linearity and provide clear interpretability, making them useful for identifying dominant factors influencing temperature. To address potential nonlinearities, intermediate-complexity models such as polynomial regression and decision trees were introduced. These models allow for curvature in relationships and can better represent interactions between spatial and temporal variables without requiring overly complex architectures.

Finally, an advanced model in the form of a neural network was implemented to capture higher-order interactions and complex synergies between features. Given the combined temporal, geographic, and urbanization data, neural networks are well-suited to model patterns that are difficult to express analytically.

Full list of models chosen for analysis:

- Null Mean Model
- Multiple Linear Regression
- Polynomial Regression 2^{nd} Order
- 2 Hidden Layer Neural Network (hidden layer sizes = [64, 32], RELU, eta = 0.001)
- Decision Tree (max depth = 12, min_samples_leaf = 30)

Explanation of Why Techniques Were Chosen

A range of models was selected to evaluate performance across increasing levels of complexity while maintaining a clear baseline for comparison. The null mean model was included as a reference point to establish the minimum expected performance, ensuring that more advanced models provide meaningful predictive improvement. Multiple linear regression and second-order polynomial regression were chosen as interpretable models to capture linear and mildly nonlinear relationships between variables such as latitude, time, and population. However, exploratory data analysis suggested weak linear relationships (e.g., low correlation between population and temperature) and more complex spatial-temporal patterns, indicating that these models would likely underperform.

To better capture these nonlinear interactions, more flexible models were introduced. A decision tree was selected for its ability to model nonlinear relationships and feature interactions without requiring explicit functional assumptions. Finally, a two-hidden-layer neural network was implemented to capture higher-order dependencies and complex synergies across temporal and geographic features. This choice was justified by the structure of the data, where patterns vary across both space and time. Consistent with expectations from EDA, the neural network demonstrated superior performance, confirming that the underlying relationships are inherently nonlinear and better suited to more expressive modeling approaches.

Feature Selection

Feature selection was performed using a subset search approach combined with time-series cross-validation, using linear regression RMSE as the evaluation metric. Importantly, this process was conducted only on the training dataset to prevent data leakage and ensure that feature choices were based solely on information available prior to model evaluation. By using a time-aware cross-validation scheme, the methodology preserves the temporal structure of the data and more accurately reflects real-world forecasting conditions.

The subset search evaluated different combinations of candidate features and ranked them based on their cross-validated RMSE. As shown in Table 6, the top-performing subsets produced nearly identical RMSE values, with the best model achieving a mean RMSE of approximately 6.51. Notably, the winning subset included all seven candidate features, indicating that each variable contributed meaningful predictive value. This result suggests that temperature patterns are influenced by a combination of spatial, temporal, and urban-related factors rather than a small isolated subset.

Table 5: Candidate Feature Flags

Feature	Selected for Final?
Population (Log Transformed)	True
Latitude	True

Longitude	True
Absolute Lattitude	True
Year	True
Month (Sine Transformed)	True
Month (Cosine Transformed)	True

Table 6: Top Cross-Validation Subsets

Rank	Feature Count	CV RMSE Mean	CV RMSE Std
1	7	6.51	0.10
2	6	6.51	0.10
3	6	6.52	0.06
4	6	6.52	0.09
5	5	6.52	0.06

The final selected features, summarized in Table 5, are: log_population, latitude, longitude, abs_latitude, year, month_sin, and month_cos.These features were chosen based on their demonstrated contribution to predictive performance. Geographic variables such as latitude, longitude, and absolute latitude capture spatial dependencies, while year accounts for long-term temporal trends such as global warming. The log transformation of population mitigates skewness caused by the concentration of large urban centers, improving numerical stability and model sensitivity. Additionally, month_sin and month_cos encode the cyclical nature of seasonal variation, allowing the model to properly represent periodic temperature behavior.

Overall, this feature selection process ensures that the final model is built on a data-driven foundation, rather than subjective assumptions. By relying on cross-validated performance metrics, the selected features can be confidently justified as those that provide the greatest predictive utility while maintaining robustness and generalizability.

Reporting and Interpretation of Results

Five predictive models were evaluated using a chronological train–test split to preserve the temporal structure of the data. The models were trained on historical observations from 1743 to 1959 and tested on data from 1960 to 2013. This approach prevents data leakage from future periods into the training set and provides a more realistic assessment of model performance in a forecasting context.

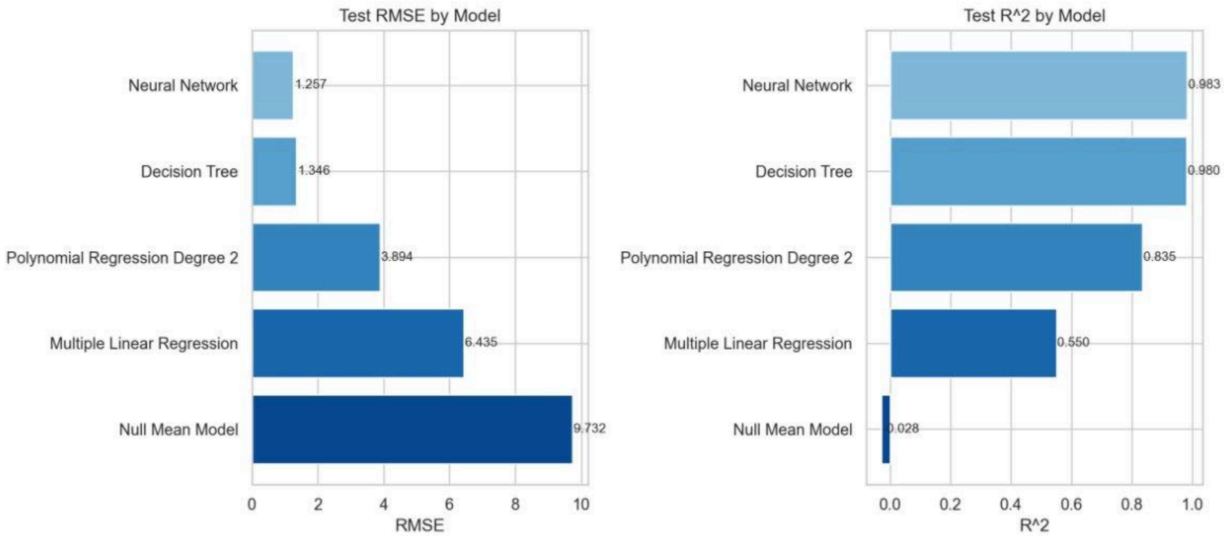

Figure 9: Model Performance Comparison

Figure 9 compares the performance of all five models using test RMSE and R^2, providing a clear ranking of predictive capability. The neural network achieved the best overall performance, with the lowest RMSE (1.26) and the highest R^2 (0.983), indicating excellent predictive accuracy and strong explanatory power. The decision tree performed similarly well, with only slightly higher RMSE (1.35) and a comparable R^2 (0.980), suggesting that nonlinear models are highly effective for this problem. In contrast, the polynomial regression model (degree 2) showed moderate performance (RMSE 3.89, R^2 0.835), capturing some nonlinearity but lacking the flexibility of more advanced models.

The multiple linear regression model performed significantly worse (RMSE 6.44, R^2 0.55), confirming earlier insights from EDA that linear relationships alone are insufficient to model temperature dynamics. As expected, the null mean model performed the worst (RMSE 9.73, $R^2 \sim 0$), serving as a baseline and demonstrating the substantial improvement gained by incorporating spatial and temporal features. Overall, these results reinforce that temperature patterns are inherently nonlinear, and more expressive models such as neural networks and decision trees are better suited to capture the complex interactions present in the data.

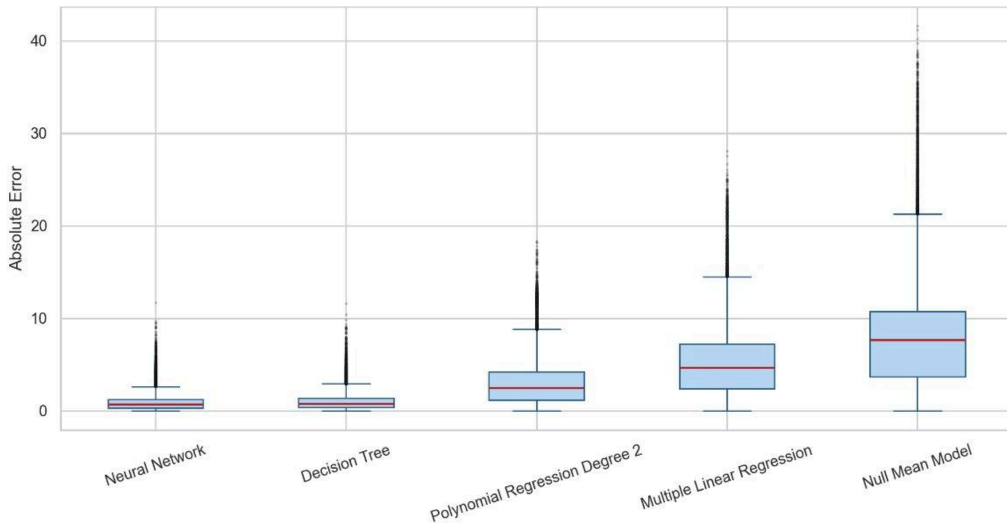

Figure 10: Absolute Error Distribution by Model

Figure 10 presents the distribution of absolute prediction errors for each model, providing deeper insight beyond aggregate metrics like RMSE. The neural network exhibits the lowest median error and the tightest interquartile range, indicating not only high accuracy but also strong consistency across predictions. The decision tree shows a very similar distribution, with slightly higher spread, confirming its competitive performance observed earlier. Both models have relatively few extreme outliers, suggesting they generalize well across different cities and time periods.

In contrast, the polynomial regression model displays a noticeably wider error distribution, with higher median error and more frequent large deviations, indicating limited ability to capture complex patterns. The multiple linear regression model performs worse, with a broader spread and larger median error, reinforcing that linear assumptions are insufficient for this dataset. The null mean model shows the highest error variability and extreme outliers, as expected from a baseline model with no feature input. Overall, this figure reinforces that nonlinear models not only reduce average error but also produce more stable and reliable predictions across the dataset.

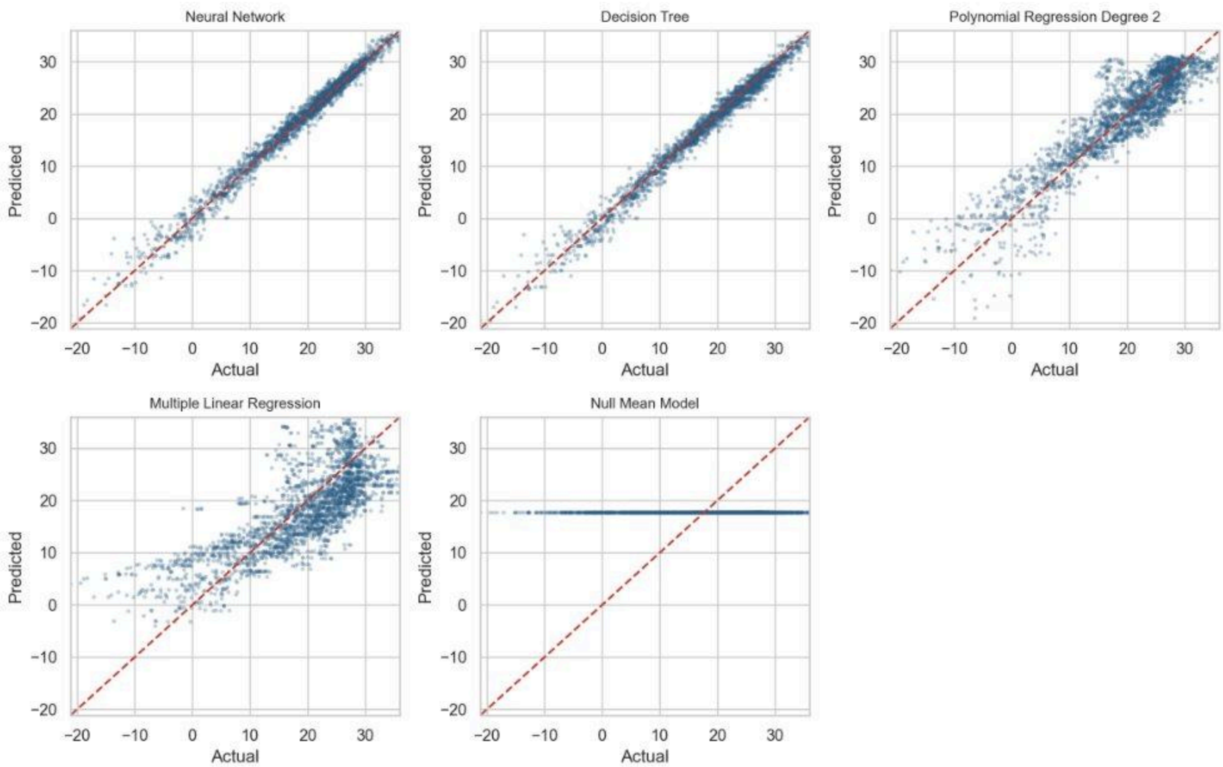

Figure 11: Scatter plots of Actual vs Predicted Temperatures

Figure 11 compares predicted versus actual temperatures for all models, providing a visual assessment of accuracy and bias. The neural network shows predictions tightly clustered along the 45-degree reference line, indicating excellent agreement with actual values and minimal systematic error. The decision tree demonstrates similarly strong performance, with only slightly more dispersion, confirming its ability to capture nonlinear relationships effectively. In contrast, the 2^{nd} order polynomial regression model exhibits noticeable spread and deviation from the diagonal, particularly at temperature extremes, suggesting limited flexibility in capturing complex patterns.

The multiple linear regression model shows even greater dispersion and clear bias, with predictions failing to align closely with actual values, especially at higher and lower temperature ranges. The null mean model performs worst, producing nearly constant predictions regardless of actual values, which results in a horizontal band and complete failure to capture variability. Overall, this figure reinforces that nonlinear models, particularly the neural network, provide the most accurate and well-calibrated predictions for this type of data, while simpler models struggle to represent the underlying dynamics.

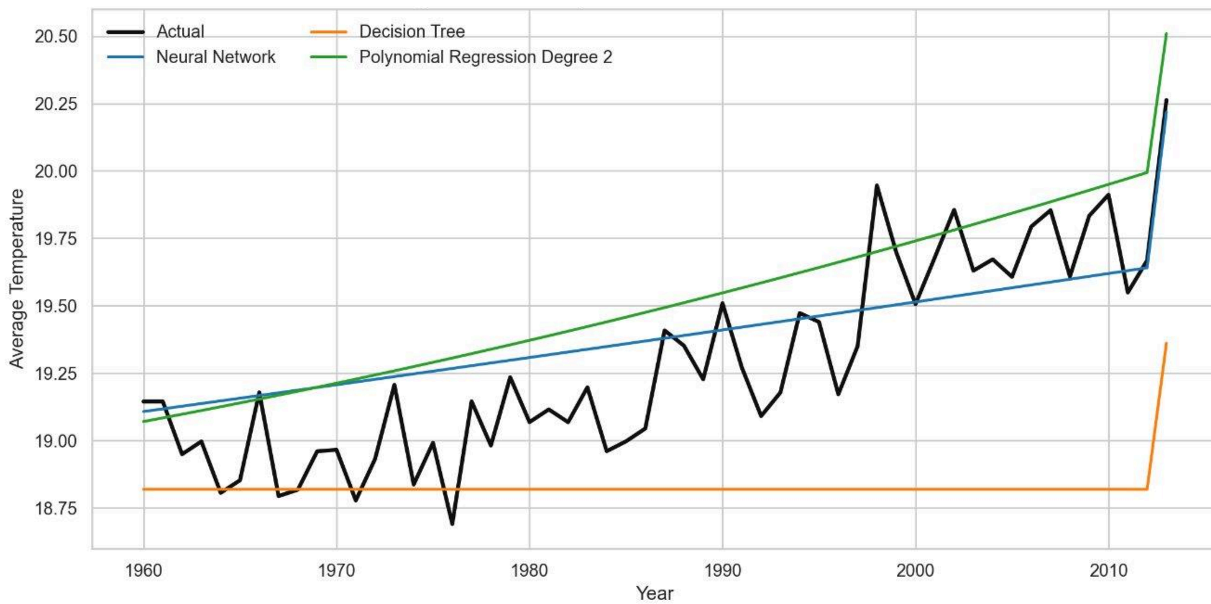

Figure 12: Average Test-Period Temperature: Actual vs Best Model Predictions

Figure 12 compares the actual average temperature trend during the test period with predictions from the top-performing models, highlighting differences in how well each captures temporal behavior. The neural network closely tracks the upward trend and short-term fluctuations in the data, showing strong alignment with the actual temperature curve. This indicates its ability to capture both long-term warming trends and local variability, making it the most accurate and responsive model. The polynomial regression model captures the general upward trend but appears overly smooth and slightly overestimates temperatures toward the end of the period, reflecting its limited flexibility in adapting to finer variations.

The decision tree, on the other hand, struggles to model the temporal progression, producing a nearly flat prediction for most of the period with a sudden adjustment at the end. This suggests poor generalization in capturing gradual trends over time. There is also a noticeable artifact near the end of the test period, where predictions (particularly from the polynomial model) spike more sharply than the actual data. This is likely due to edge effects or limited training data near the boundary, causing extrapolation behavior that deviates from the true trend. Overall, this figure reinforces that while simpler nonlinear models can approximate broad trends, the neural network provides the best balance between trend tracking and responsiveness to fluctuations, making it the most suitable model for this dataset.

Table 7: Model metrics

model	train start year	train end year	test start year	test end year	train rows	test rows	RMSE	R^2
4L Neural Network	1743	1959	1960	2013	163768	64407	1.256882	0.982854
Decision Tree	1743	1959	1960	2013	163768	64407	1.346274	0.980328
Polynomial Regression (2^{nd} Order)	1743	1959	1960	2013	163768	64407	3.894094	0.835416
Multiple Linear Regression	1743	1959	1960	2013	163768	64407	6.435498	0.550489
Null Mean Model	1743	1959	1960	2013	163768	64407	9.732317	-0.02804

Discussions

The neural network emerged as the strongest overall model, achieving the lowest error and highest explanatory power. This indicates that average temperature is highly predictable from the selected spatial and temporal features when nonlinear relationships are properly captured. The strong performance of both the neural network and decision tree further reinforces that the underlying structure of the data is nonlinear and interaction-driven, rather than purely linear.

From exploratory analysis and modeling results, latitude exhibits a much stronger relationship with temperature (-0.393) compared to log population (-0.030). This confirms that geographic location is the dominant driver of temperature variation across cities. Additionally, year shows a positive association with temperature (0.147), which aligns with the observed long-term warming trend and is consistent with broader climate research. Improvements in data quality over time were also evident, as shown by the strong negative correlation between temperature uncertainty and year, indicating more reliable measurements in recent periods.

Conclusion

One key limitation of this study is that the population dataset is effectively static and modern, meaning it does not capture how cities have grown over time. As a result, population serves primarily as a proxy for city scale, rather than a dynamic variable capable of explaining long-term temperature changes. This limits the ability to draw causal conclusions about urbanization effects, such as the urban heat island phenomenon, over historical periods.

Overall, the results support a predictive interpretation more strongly than a causal one. While the models can accurately estimate temperature based on spatial and temporal inputs, the weak relationship between population and temperature suggests that urbanization alone is not a dominant global driver in this dataset. Instead, geographic and temporal factors.

particularly latitude and long-term trends, play the primary roles. Additionally, the consistency between EDA findings and model performance strengthens confidence in the results. Simpler models struggled due to their inability to capture nonlinear structure, while more flexible models succeeded, confirming that temperature dynamics involve complex interactions across space and time. Minor artifacts observed at the edges of the test period also highlight the challenges of extrapolation in time-series data, suggesting that future work could benefit from more temporally complete auxiliary datasets.

In summary, this project demonstrates that while machine learning models can effectively predict temperature patterns, careful consideration must be given to data limitations and variable interpretation when making broader claims about causality.

Recommendations of Study and Future Research

While our current study demonstrated a strong predictive capacity of neural networks and decision trees for modeling urban temperature patterns, there are a few key implementations we can analyze to substantially deepen this line of research.

One promising next step would be the incorporation of gradient boosting algorithms such as XGBoost or LightGBM into the modeling pipeline. Gradient boosting methods build sequential decision trees, each correcting the residuals of the last, and minimizing the loss function of the model as a result. Given the mixed temporal and spatial nature of our datasets, gradient boosting would likely capture interaction effects between features like latitude, year, and log population more efficiently while offering greater interpretability through built-in feature importance scores which would allow us to more precisely quantify the relative contributions of urban factors to temperature variation.

Another way we can improve our study is by narrowing the geographic scope of the analysis to a single, well-documented metropolitan area such as New York City. The current study spans 100 cities across 49 countries, which introduces climatic heterogeneity. For instance, cities may be driven by entirely different temperature regimes, making it difficult to isolate the specific influence of urbanization on local warming. By concentrating on one city over a long time period before applying the corresponding model, future work can leverage finer-tuned data such as neighborhood-level census records, historical building density maps, and surface albedo measurements to build a more detailed model of how urban growth directly drives heat accumulation.

Finally, the integration of vegetation and land-use variables in future research would help address critical gaps in our existing research. The current feature set does not include any measure of green land areas, yet vegetation is one of the most well-established mitigating factors of urban heat, as trees and plants reduce surface temperatures through evapotranspiration and shading. Incorporating indices such as the Normalized Difference Vegetation Index (NDVI), derived from satellite imagery, alongside land-use classifications would allow future models to capture not only where heat is accumulating but why, and what targeted urban greening interventions such as expanding parks, planting street trees, or developing green rooftops could do to reverse current warming trends. Throughout this process, distinguishing between impervious surfaces, parks, and residential areas versus commercial zones is also critical.

Overall, adopting gradient boosting methods, focusing on a centralized urban case study like New York City, and incorporating vegetation and land-use data would move this research further to help us understand and work towards mitigating the urban heat island effect.

Appendix

Appendix 1. Numeric Summary of Features

	count	mean	std	min	25%	50%	75%	max
average temperature	228175	18.12597	10.0248	-26.772	12.71	20.428	25.918	38.283
temperature uncertainty	228175	0.969343	0.979644	0.04	0.34	0.592	1.32	14.037
population	228175	5905357	4070569	1200000	2746388	4591246	8443675	22315474
log population	228175	15.3801	0.643837	13.99783	14.8258	15.33966	15.94893	16.92079
latitude	228175	22.9092	22.01098	-37.78	13.66	28.13	39.38	60.27
longitude	228175	43.99105	65.29996	-118.7	4.05	45	103.66	151.78
year	228175	1913.893	62.02598	1743	1869	1918	1966	2013
month	228175	6.494761	3.451441	1	3	6	9	12

Appendix 2: Model feature types

feature	feature_type
log_population	numeric
latitude	numeric
longitude	numeric
abs_latitude	numeric
year	numeric
month_sin	numeric
month_cos	numeric

Appendix 3: Missing Values after Pre-Processing

	missing_values
date	0
year	0
month	0
decade	0
city	0
country	0
latitude	0
longitude	0
abs_latitude	0
population	0
log_population	0
month_sin	0
month_cos	0
average_temperature	0
temperature_uncertainty	0
temp_change_from_city_mean	0
matched_population_city	0
matched_population_country	0
feature_code	0

Appendix 4: Summary of Cities

city	country	records	Avg_temperature [C]	population	latitude
Umm Durman	Sudan	1768	29.08129	1200000	15.27
Madras	India	2508	28.41786	4681087	13.66
Jiddah	Saudi Arabia	1848	27.69207	2867446	21.7
Ho Chi Minh City	Vietnam	2069	27.19398	8993082	10.45
Bangkok	Thailand	2246	27.16473	5104476	13.66
Mogadishu	Somalia	1679	27.15196	2587183	2.41
Fortaleza	Brazil	1659	27.00864	2400000	-4.02
Hyderabad	India	2508	26.86933	6809970	16.87
Surabaya	Indonesia	1879	26.8099	2874314	-7.23
Rangoon	Burma	2532	26.73519	4477638	16.87
Bombay	India	2508	26.63145	12691836	18.48
Lagos	Nigeria	1758	26.56593	15388000	5.63
Ahmadabad	India	2448	26.52985	6357693	23.31
Singapore	Singapore	2052	26.5231	5638700	0.8
Jakarta	Indonesia	1879	26.50821	8540121	-5.63
Manila	Philippines	2021	26.44833	1600000	15.27
Ibadan	Nigeria	1721	26.37311	3649000	7.23
Surat	India	2508	26.33086	4591246	21.7
Abidjan	CÃ´te D'Ivoire	1777	26.16374	3677115	5.63
Kano	Nigeria	1721	26.14054	4103000	12.05
Calcutta	India	2532	26.04215	4631392	23.31
Karachi	Pakistan	2324	26.01726	11624219	24.92
Santo Domingo	Dominican Republic	2207	25.97678	2201941	18.48
Dar Es Salaam	Tanzania	1586	25.74467	2698652	-7.23
Nagpur	India	2508	25.65502	2228018	21.7
Dhaka	Bangladesh	2532	25.49057	10356500	23.31

Jaipur	India	2399	25.39306	2711758	26.52
Riyadh	Saudi Arabia	2008	25.26333	4205961	24.92
New Delhi	India	2394	25.16586	10927986	28.13
Delhi	India	2394	25.16586	10927986	28.13
Bangalore	India	2508	24.8559	8443675	12.05
Kanpur	India	2452	24.76004	2823249	26.52
Lakhnau	India	2452	24.76004	2472011	26.52
Salvador	Brazil	2096	24.65697	2711840	-13.66
Pune	India	2508	24.64462	3124458	18.48
Dakar	Senegal	1872	24.62752	2476400	15.27
Lahore	Pakistan	2083	24.1384	6310888	31.35
Faisalabad	Pakistan	2083	24.1384	2506595	31.35
Kinshasa	Congo (DR)	1735	23.86644	7785965	-4.02
Rio De Janeiro	Brazil	2096	23.78892	6023699	-23.31
Luanda	Angola	1735	23.69305	2776168	-8.84
Baghdad	Iraq	2320	22.61435	7216000	32.95
Taipei	Taiwan	2071	22.21244	7871900	24.92
Cali	Colombia	1985	21.79703	2392877	4.02
BrasÃlia	Brazil	2096	21.72759	2207718	-15.27
Guangzhou	China	2084	21.60868	16096724	23.31
Cairo	Egypt	2442	21.22126	9606916	29.74
Gizeh	Egypt	2442	21.22126	4367343	29.74
Belo Horizonte	Brazil	2096	21.0714	2373224	-20.09
Durban	South Africa	1873	20.35129	3120282	-29.74
Alexandria	Egypt	2666	20.31262	5263542	31.35
Harare	Zimbabwe	1690	20.20784	1542813	-18.48
BogotÃ¡	Colombia	2018	20.00226	7674366	4.02
SÃ£o Paulo	Brazil	2096	19.69937	10021295	-23.31

Addis Abeba	Ethiopia	1679	17.52507	2757729	8.84
Aleppo	Syria	2479	17.37059	1602264	36.17
Izmir	Turkey	3128	17.27806	2500603	37.78
Casablanca	Morocco	2914	17.18416	3144909	32.95
Sydney	Australia	2071	17.0023	4627345	-34.56
Chongqing	China	2083	16.84496	7457599	29.74
Wuhan	China	2072	16.83094	10392693	29.74
Lima	Peru	1521	16.76912	7737002	-12.05
Nairobi	Kenya	1678	16.0814	2750547	-0.8
Cape Town	South Africa	1873	16.07908	4710000	-32.95
Shanghai	China	2072	15.96951	22315474	31.35
Los Angeles	United States	1977	15.87804	3898747	34.56
Mexico	Mexico	2145	15.71742	12294193	20.09
Nanjing	China	2072	15.60205	9314685	31.35
Nagoya	Japan	2030	14.75507	2191279	34.56
Kabul	Afghanistan	2121	14.34292	4434550	34.56
Istanbul	Turkey	3166	13.50741	14804116	40.99
Melbourne	Australia	2067	13.3743	4246375	-37.78
Jinan	China	2084	13.08981	4335989	36.17
Mashhad	Iran	2155	12.57199	2307177	36.17
Tokyo	Japan	2020	12.556	8336599	36.17
Rome	Italy	3166	11.9655	2318895	42.59
Peking	China	2304	11.80938	18960744	39.38
Tianjin	China	2304	11.80938	11090314	39.38
Tangshan	China	2290	11.57937	3372102	37.78
Xian	China	2126	11.48715	9600000	34.56
Madrid	Spain	3166	11.4487	3255944	40.99
Seoul	South Korea	2096	10.68188	10349312	37.78

Chengdu	China	2183	10.63804	13568357	31.35
Paris	France	3166	10.40264	2138551	49.03
Ankara	Turkey	2945	10.39212	3517182	39.38
Dalian	China	2135	10.22908	4913879	39.38
Chicago	United States	3141	10.07064	2746388	42.59
New York	United States	3119	9.523296	8804190	40.99
London	UK	3166	9.459038	8961989	52.24
Berlin	Germany	3166	8.916234	3426354	52.24
Taiyuan	China	2311	7.976806	4303673	37.78
Shenyang	China	2199	7.21498	7050000	40.99
Kiev	Ukraine	3166	7.041033	2797553	50.63
Toronto	Canada	3141	5.773911	2600000	44.2
Santiago	Chile	1900	5.692277	4837295	-32.95
Changchun	China	2316	4.923798	4714996	44.2
Montreal	Canada	3141	4.445014	1762949	45.81
Moscow	Russia	3166	3.999711	10381222	55.45
Saint Petersburg	Russia	3166	3.918045	5351935	60.27
Harbin	China	2317	3.625744	5242897	45.81